现代 LaTeX 入门讲座

Modern LaTeX in a Nutshell



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"Your paper makes no goddamn sense,
but it's the most beautiful thing
I have ever laid eyes on."

From r/ProgrammerHumor 

介绍

历史回眸



高德纳（Donald E. Knuth）

TEX



Leslie Lamport

LATEX

图片来源：

LaTeX 是什么?

- 发音：
  - /ˈlɑːtɛx/ or /ˈleɪtɛx/ or whatever you like
- 打公式方便?
  - 「复杂公式输入哪家强，当然首选 LaTeX 帮忙」
- 写论文神器?
  - 「想要轻松给论文排版，当然少不了 LaTeX 啦」
- 不想做宏编程的标记语言不是好的排版引擎?
  - LaTeX is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation. LaTeX is the *de facto* standard for the communication and publication of scientific documents. LaTeX is available as free software. 🔗

LaTeX 是什么？——What you *think* is what you get!

```
\documentclass{article}
\usepackage{amsmath,graphicx}
\title{Normal distribution}
\author{Wikipedia, the free encyclopedia}

\begin{document}
\maketitle
\section{Introduction}
% 省略一些内容……
The probability density of the normal
distribution is
\begin{equation}
  f(x|\mu, \sigma)
  = \frac{1}{\sqrt{2\pi\sigma^2}}
    e^{-\frac{(x-\mu)^2}{2\sigma^2}}
\end{equation}
where
\begin{itemize}
  \item $\mu$ is the mean of the distribution
  \item $\sigma$ is the standard deviation
\end{itemize}
\end{document}
```

来源：Wikipedia 🔗

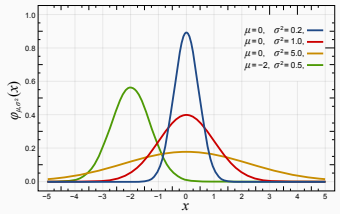
Normal distribution

Wikipedia, the free encyclopedia

March 12, 2022

1 Introduction

In probability theory, the normal distribution is a very common continuous probability distribution. Normal distributions are important in statistics and are often used in the natural and social sciences to represent real-valued random variables whose distributions are not known. A random variable with a Gaussian distribution is said to be normally distributed and is called a normal deviate.



The probability density of the normal distribution is

$$f(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{1}$$

where

- μ is the mean of the distribution
- σ is the standard deviation

基本原则

- 排版 vs 文字处理
  - 《别把 LaTeX 当 Word 用》
- 遵循业（xué）界（xiào）规范
  - 《管教务处 or 研究生院 or 物理系叫爸爸》
- 追求良好的阅读体验（readability）
- 内容与格式分离
- **内容永远比格式重要！**
  - *Typography exists to honor content.* —R. Bringhurst

安装

不想安装?

- 云端服务可能更好用
- 免去安装、升级等一系列烦恼，可以多人协作
- 国际版：ShareLaTeX 和 Overleaf（现已合并）
  - 模板丰富
  - 用户支持很好（支持团队中有华人）
  - 注册及使用可能遇到网络问题
- 国内版：TeXPage 🔗
  - 网络限制较少
  - 支持更多的中文字体
  - 不够成熟稳定
  - 免费账号项目数量受限

选择发行版

- TeX 发行版（distribution）
  - 引擎、宏包、字体、文档的综合体
  - 类比 Visual Studio
  - TeX Live、MacTeX、MiKTeX 等
- TeX Live 🔗
  - 官方维护，首选，跨平台
  - MacTeX ≈ macOS 下的 TeX Live
  - 缺点：完整版体积大（3GB+）、每年需重装
- MiKTeX 🔗
  - 由 Christian Schenk 维护（是个狠人）
  - 宏包随用随装
  - 缺点：部分细节与 TeX Live 不兼容、网络问题
- **不要安装 CTeX 套装！**
  - **存在严重 bug，并且完全过时**

下载

- 选择国内 CTAN 镜像
  - 清华大学开源软件镜像站
  - 上海交通大学软件源镜像服务
  - 中国科学技术大学开源软件镜像
  - 复旦大学……
- 建议使用 ISO 镜像离线安装
- 在线安装要求网络稳定

安装流程

- 新手建议安装完整版 TeX Live 或 MacTeX
  - 完整版可避免后续维护烦恼
  - 一路点击「下一步」
  - 保持耐心，做好重装的打算
- Linux specials
  - 软件源更新较慢，可以考虑 Vanilla TeX Live
  - GUI 安装界面需要 `perl-tk` 等
  - 环境变量、`fontconfig`、dummy package 配置
- macOS specials
  - 推荐 Homebrew
- 手把手的教程：

神圣的战争——选择编辑器

- 专用型
  - TeXworks：TeX Live 自带
  - TeXstudio：功能丰富，对新手友好
  - TeXShop：MacTeX 自带
  - WinEdt：功能丰富，收费
- 通用型
  - Visual Studio Code：配合 LaTeX Workshop 插件
  - Sublime Text：需要收费
  - Vim：`q`、`q!`、`wq`、`wq!`
- 编辑器对比：🔗 🔗 🔗

开始之前……

命令行基础

- 打开终端
  - : 右键开始菜单、空白处 ⇧ + 右键、 + R & `cmd`
  - : Ctrl + Alt + T
  - : ⌘ + 空格 搜索 Terminal、可在 Finder 中添加服务
- 基本命令：
  - `cd`、`ls/dir`、`rm/del`、`clear/cls`
  - 选项：`-h`、`--help`、`/?`
- 其他：
  - 复制粘贴：Ctrl/⇧ + Ins、Ctrl/⌘ + C/V、
  - 路径连接符：斜线（`/`）或反斜线（`\`）
  - 换行符：LF（`\n`）或 CRLF（`\r\n`）
  - 结束进程：Ctrl + C
- **尽量不要用中文；避免空格、特殊符号**

编码

关于 LaTeX 源文件的编码，我们给出如下结论：

编码定理

一般地，在任何场合使用（不带 BOM 的）**UTF-8** 编码均是最优选择.

此定理的证明留做习题.

填写内容

Hello world!

```latex
% 用 pdfLaTeX、XeLaTeX 或 LuaLaTeX 编译
\documentclass{article}
\begin{document}
Hello world!
\end{document}

% 用 XeLaTeX 或 LuaLaTeX 编译
\documentclass{ctexart}
\begin{document}
你好，世界！
\end{document}
```

引擎与格式

- **引擎**：TeX 的实现
  - pdfTeX：直接生成 PDF，支持 micro-typography
  - XeTeX：支持 Unicode、OpenType 与复杂文字编排（CTL）
  - LuaTeX：支持 Unicode、OpenType，内联 Lua
  - (u)pTeX：日本方面推动，生成 `.dvi`，（支持 Unicode）
  - ApTeX：底层 CJK 支持，内联 Ruby，Color Emoji（手动斜眼笑）
- **格式**：TeX 的语言扩展（命令封装）
  - plain TeX：Knuth 同志专用
  - LaTeX：排版科技类文章的事实（*de facto*）标准
  - ConTeXt：基于 LuaTeX 实现，优雅、易用（吗?）
- **程序**：引擎 + dump 之后的格式代码
  - **英文文章：pdfLaTeX、XeLaTeX 或 LuaLaTeX**
  - **中文文章：XeLaTeX 或 LuaLaTeX**

编译

- 现代 TEX 引擎均可直接生成 PDF
- 命令行
  - `pdflatex`/`xelatex`/`lualatex` + `<文件名>[.tex]`
  - 多次编译：读取并排版中间文件
  - 推荐 `latexmk`：`latexmk [<选项>] [<文件名>]`
- 编辑器
  - 按钮的背后仍然是命令
  - `PATH` 环境变量：确定可执行文件的位置
  - VS Code：配置 `tools` 和 `recipes`

语法

- 注释以 `%` 开头，忽略其后所有内容
- 命令以 `\` 开头，区分大小写
  - `\foo{arg}`：必选参数放在 `{...}` 中
  - `\foo[bar]{arg}`：可选参数放在 `[...]`
- 环境

```
\begin{env}
  ...
\end{env}
```

- 特殊符号需要转义：`\%`、`\$`、`\&`、`\textbackslash` 等
- 连续多个空格 = 单个空格 = 单个换行符
- TEX/LATEX 的语法可以修改

文件结构

```latex
% 用 UTF-8 编码，命名为 xxx.tex
\documentclass{article}                          % 指明文档类型：文章
% 导言区：设置文档样式
\usepackage{amsmath}                             % 调用宏包，实现各种功能
\newcommand{\keyword}[1]{\textbf{#1}}            % 自定义命令

\begin{document}
% 正文：套用格式
In quantum mechanics, the \keyword{Schr\"odinger equation} is a
mathematical equation that describes the changes over time of a
physical system in which quantum effects, such as \keyword{wave--%
particle duality}, are significant.

% 上面的空行表示分段
In classical mechanics, Newton's second law
($\boldsymbol{F}=m\boldsymbol{a}$) is used to make a\ldots{}

Time-dependent Schrödinger equation can be written as  % ö 也能直接用
\[ i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle. \]
\end{document}
```

Schrödinger equation

In quantum mechanics, the **Schrödinger equation** is a mathematical equation that describes the changes over time of a physical system in which quantum effects, such as **wave–particle duality**, are significant.

In classical mechanics, Newton's second law ($F = m\boldsymbol{a}$) is used to make a...

Time-dependent Schrödinger equation can be written as

$$i\hbar\frac{d}{dt}|\Psi(t)\rangle = \hat{H}\,|\Psi(t)\rangle.$$

谋篇布局

- 文档部件
  - 标题：`\title`、`\author`、`\date` → `\maketitle`
  - 摘要：`abstract` 环境
  - 目录：`\tableofcontents`
  - 章节：`\chapter`、`\section`、`\subsection` 等
  - 文献：`\bibliography`
- 文档划分
  - 凤头猪肚豹尾：`\frontmatter`、`\mainmatter`、`\backmatter`
  - 分文件编译：`\include`、`\input`

文本标记（一）

- 加粗：`{\bfseries ...}` 或 `\textbf{...}`
- 倾斜：`{\itshape ...}` 或 `\textit{...}`
- 字号：`\tiny`、`\small`、`\large`、`\Large` 等
- 换行：`\\`
- 缩进：`\indent`
- 居中：`\centering` 或 `center` 环境

请忘记上一页

文本标记（二）

- 为什么要有不同的标记？——表达不同的**语义**
- `\textbf` 这样的命令是否表达语义？
- 再提一遍基本原则：**内容与格式分离**
- 正确（或曰：合理）的做法
  - 强调文字（意大利体）：`\emph{...}`
  - 摘要（居中，小字号，带有标题）：`abstract` 环境
  - 引用（左右边距较大）：`quote` 或 `quotation` 环境
  - 自定义新的命令、环境
- 报告，我想偷懒！
- 报告，我听说了 Tailwind CSS 🔗！

常用环境：列表与枚举

```
\begin{enumerate}
  \item Frontend
    \begin{itemize}
      \item React
      \item Vue.js
      \item Svelte
    \end{itemize}
  \item Backend
    \begin{description}
      \item[PHP] Laravel
      \item[JavaScript] Express
      \item[Python] Django
    \end{description}
\end{enumerate}
```

1. Frontend
   - React
   - Vue.js
   - Svelte
2. Backend

   PHP Laravel
   JavaScript Express
   Python Django

常用环境：图片

```latex
% 不是 graphics
\usepackage{graphicx}
% 可以统一指定图片路径
\graphicspath{{./images/}}

\begin{figure}
  \centering
  % 可指定宽度、高度等选项
  % 图片后缀名可省略，但建议保留
  \includegraphics[...]{fudan-logo.pdf}
  \caption{Logo of Fudan University}
  \label{fig:fudan-logo}
\end{figure}
```



Figure 3: Logo of Fudan University

常用环境：表格

```
\usepackage{booktabs} % 三线表
\begin{table}
  \caption{Population Census of China}
  \label{tab:china-population}
  % 列格式：c 居中，l 左对齐，r 右对齐
  \begin{tabular}{cc}
    \toprule
      Year & Population \\
    \midrule
      1953 &  6.0 \\
      ...
      2020 & 14.1 \\
    \bottomrule
  \end{tabular}
\end{table}
```

Table 1: Population Census of China

Year	Population
1953	6.0
1964	6.9
1982	10.1
1990	11.3
2000	12.7
2010	13.4
2020	14.1

常用环境：定理

```
\usepackage{amsthm}
% 需要预先定义
\newtheorem{theorem}{Theorem}
\newtheorem*{remark}{Remark} % 不编号

\begin{theorem}[Fermat]
  $a^n+b^n=c^n$ has no positive...
\end{theorem}
\begin{proof}
  % 证明后面会有 QED 符号
  It's obvious.
\end{proof}
\begin{remark}
  The cases $n=1$ and $n=2$...
\end{remark}
```

Theorem 1 (Fermat). *$a^n + b^n = c^n$ has no positive integer solutions for x, y and z when n is greater than 2.*

Proof. It's obvious. □

Remark. *The cases $n = 1$ and $n = 2$ have been known since antiquity to have infinitely many solutions.*

浮动体与交叉引用

- 浮动体机制
  - `figure` 和 `table` 环境，标题使用 `\caption` 命令
  - 位置控制：`\begin{figure}[htb]`
  - 希望浮动体不要乱跑：「这是病，得治」
  - 文本为主，图、表为辅
  - 避免「见上图」、「见下表」
  - 建议写完全文之后统一调整
- 以标签控制交叉引用
  - 被引处：`\label`
  - 引用处：`\ref`、`\eqref` 等（如图 3、表 1）
  - 用有意义的标签：`\label{eq:euler-lagrange-eq}`
  - 需多次编译——推荐 `latexmk`

如何在论文中画出漂亮的插图?

- 外部插入
  - Mathematica、MATLAB
  - PowerPoint、Visio、Adobe Illustrator、Inkscape、Figma 等
  - Python `Matplotlib`、`Plots.jl`、R、Plotly 等
  - draw.io 🔗、Mathcha 🔗、ProcessOn 🔗 等网站
- TEX 内联
  - Asymptote
  - **`pgf/TikZ`、`pgfplots`**
- 插图格式
  - 矢量图：`.pdf`
  - 位图：`.jpg` 或 `.png`
  - **不再推荐 `.eps`**
  - 不（完全）支持 `.svg`、`.bmp`
- 参考：🔗 🔗 🔗

文献

- 建议自动生成（你只有三篇参考文献?）
- `.bib` 数据库（条目会包含 key，用于引用）
  - Google Scholar 复制，Zotero、Jabref 等生成
  - 注意特殊符号、公式等常常需要人工检查
- 传统方法：BibTeX 后端
  - 指定样式：`\bibliographystyle{<style>}`（导言区）
  - 标记引用：`\cite{<key>}`
  - 插入参考文献：`\bibliography{<bib 文件>}`
  - 更多文献、引用样式：`natbib` 宏包
  - 国家标准 GB/T 7714–2015 🔗🔗：**`gbt7714` 宏包**
- 现代方法：`biber` 后端 + `biblatex` 宏包
  - 国家标准：`biblatex-gb7714-2015` 宏包
- 需多次编译——再次推荐 `latexmk`



U+30EDE

中文支持

- 中文有什么特殊?
  - 汉字太多（97,057+）
  - 横排 + 直排、标点禁则、行间注 🔗
  - にほんご、देवनागरी、العَرَبِيَّة、 
- 已淘汰：
  - CCT 系统、CJK 宏包（裸用）
  - CTEX 套装
- 目前推荐手段：
  - **ctex 宏集**（此 ctex 非彼 CTEX）
  - XELATEX 编译
- 可以用，不推荐：
  - xeCJK 宏包（裸用）
  - ctex 宏集 + 其他引擎编译

图片来源：GitHub、Twitter、知乎


刘海洋


李清


吴凌云


江疆


马起园


黄晨成


鲁尚文


李泽平


周宇恺


张瑞熹

幻灯片

- 基本框架
  - beamer 或 ctexbeamer 文档类
  - 页面由 frame 环境组织
  - 文本内容：建议使用 itemize 和 enumerate
  - 图表：不再浮动，不建议使用交叉引用
  - 定理及强调：theorem、proof、block 等
  - 分栏：columns + column 环境
- 主题与样式
  - \usetheme、\use[font|color|inner|outer]theme
  - 更现代的主题：metropolis
  - 使用「默认」字体：\usefonttheme{serif}
- 动画（覆盖）
  - \pause 命令
  - \onslide<1>、\item<1-> 等

使用模板

模板

- 是什么?
  - 设计好的格式框架
  - 专注于内容：**不必追求与期刊排版完全一致**
  - Word 中的样式：「学好 LaTeX 可以更科学地使用 Word」
- 有哪些?
  - 期刊：`revtex`、`elsarticle`、`IEEEtran`……
  - 学位论文：`thuthesis`、`ustcthesis`、`SJTUThesis`……
- 怎么用?
  - `\documentclass{...}`，配置参数，照常编写
  - 可能与 LaTeX 通常用法不同：**看文档，看文档，看文档**
- 去哪里找?
  - CTAN 或 GitHub
  - 期刊官网
  - 「湿兄用 U 盘 or 微信传给你的模板几乎一定是过时的」

revtex

- 获取
  - APS 官网 🔗
  - TeX Live 自带（注意检查版本）
  - 阅读文档：`textdoc aps`
- 写作
  - `\documentclass[aps,prl,twocolumn,...]{revtex4-2}`
  - 作者信息：`\title`、`\author` 等需放在 `\begin{document}` 之后
  - 跨栏长公式：`widetext` 环境
  - 文献引用：`\cite` + `\bibliography`，无需 `\bibliographystyle`
- 编译
  - 推荐 pdfLaTeX，也可 `latexmk -pdf`

revtex：示例



Manuscript Title:
with Forced Linebreak*

Ann Author† and Second Author‡
Authors' institution and/or address
*This line break forced with *
(MUSO Collaboration)

Charlie Author§
Second institution and/or address
This line break forced and
Third institution, the second for Charlie Author

Delta Author
Authors' institution and/or address
*This line break forced with *
(CLEO Collaboration)
(Dated: December 27, 2018)

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

Usage: Secondary publications and information retrieval purposes.

Structure: You may use the `description` environment to structure your abstract; use the optional argument of the `\item` command to give the category of each item.

I. FIRST-LEVEL HEADING: THE LINE BREAK WAS FORCED via \\

This sample document demonstrates proper use of REVTEX 4.2 (and LATEX 2_ε) in manuscripts prepared for submission to APS journals. Further information can be found in the REVTEX 4.2 documentation included in the distribution or available at `http://journals.aps.org/revtex/`.

When commands are referred to in this example file, they are always shown with their required arguments, using normal TEX format. In this format, `#1`, `#2`, etc. stand for required author-supplied arguments to commands. For example, in `\section{#1}` the `#1` stands for the title text of the author's section heading, and in `\title{#1}` the `#1` stands for the title text of the paper.

Line breaks in section headings at all levels can be introduced using \\. A blank input line tells TEX that the paragraph has ended. Note that top-level section headings are automatically uppercased. If a specific letter or word should appear in lowercase instead, you must escape it using `\lowercase{#1}` as in the word "via" above.

* A footnote to the article title
† Also at Physics Department, XYZ University.
‡ Second.Author@institution.edu
§ http://www.Second.institution.edu/~Charlie.Author

A. Second-level heading: Formatting

This file may be formatted in either the `preprint` or `reprint` style. `reprint` format mimics final journal output. Either format may be used for submission purposes. `letter` sized paper should be used when submitting to APS journals.

1. Wide text (A level-3 head)

The `widetext` environment will make the text the width of the full page, as on page 4. (Note the use the `\pagerefl{#1}` command to refer to the page number.)

a. Note (Fourth-level head is run in) The width-changing commands only take effect in two-column formatting. There is no effect if text is in a single column.

B. Citations and References

A citation in text uses the command `\cite{#1}` or `\onlinecite{#1}` and refers to an entry in the bibliography. An entry in the bibliography is a reference to another document.

1. Citations

Because REVTEX uses the `natbib` package of Patrick Daly, the entire repertoire of commands in that package

are available for your document; see the `natbib` documentation for further details. Please note that REVTEX requires version 8.31a or later of `natbib`.

a. Syntax The argument of `\cite` may be a single *key*, or may consist of a comma-separated list of keys. The citation *key* may contain letters, numbers, the dash (-) character, or the period (.) character. New with natbib 8.3 is an extension to the syntax that allows for a star (*) form and two optional arguments on the citation key itself. The syntax of the `\cite` command is thus (informally stated)

```
\cite{ key }, or
\cite{ optarg+key }, or
\cite{ optarg+key , optarg+key,... },
```

where *optarg+key* signifies

```
key, or
*key, or
[pre]key, or
[pre][post]key, or even
*[pre][post]key
```

where *pre* and *post* is whatever text you wish to place at the beginning and end, respectively, of the bibliographic reference (see Ref. [1] and the two under Ref. [2]). (Keep in mind that no automatic space or punctuation is applied.) It is highly recommended that you put the entire *pre* or *post* portion within its own set of braces, for example: `\cite{ {[{cf.}}]key}`. The extra set of braces will keep LATEX out of trouble if your *post* contains the comma (,) character.

The star (*) modifier to the *key* signifies that the reference is to be merged with the previous reference into a single bibliographic entry, a common idiom in APS and AIP articles (see below, Ref. [2]). When references are merged in this way, they are separated by a semicolon instead of the period (full-stop) that would otherwise appear.

b. Eliding repeated information When a reference is merged, some of its fields may be elided: for example, when the author matches that of the previous reference, it is omitted. If both author and journal match, both are omitted. If the journal matches, but the author does not, the journal is replaced by *ibid.*, as exemplified by Ref. [2]. These rules embody common editorial practice in APS and AIP journals and will only be in effect if the markup features of the APS and AIP BibTEX styles is employed.

c. The options of the cite command itself Please note that optional arguments to the *key* change the reference in the bibliography, not the citation in the body of the document. For the latter, use the optional arguments of the `\cite` command itself: `\cite *[pre-cite][post-cite]{key-list}`.

2. Example citations

By default, citations are numerical[3]. Author-year citations are used when the journal is RMP. To give a textual citation, use `\onlinecite{#1}`: Refs. 1 and 4. By default, the `natbib` package automatically sorts your citations into numerical order and "compresses" runs of three or more consecutive numerical citations. REVTEX provides the ability to automatically change the punctuation when switching between journal styles that provide citations in square brackets and those that use a superscript style instead. This is done through the `citesuperscript` option. For instance, the journal style `prb` automatically invokes this option because *Physical Review B* uses superscript-style citations. The effect is to move the punctuation, which normally comes after a citation in square brackets, to its proper position before the superscript. To illustrate, we cite several together [1, 2, 4–6], and once again in different order (Refs. [1, 2, 4–6]). Note that the citations were both compressed and sorted. Furthermore, running this sample file under the `prb` option will move the punctuation to the correct place.

When the `prb` class option is used, the `\cite{#1}` command displays the reference's number as a superscript rather than in square brackets. Note that the location of the `\cite{#1}` command should be adjusted for the reference style: the superscript references in `prb` style must appear after punctuation; otherwise the reference must appear before any punctuation. This example was written for the regular (non-`prb`) citation style. The command `\onlinecite{#1}` in the `prb` style also displays the reference on the baseline.

3. References

A reference in the bibliography is specified by a `\bibitem{#1}` command with the same argument as the `\cite{#1}` command. `\bibitem{#1}` commands may be crafted by hand or, preferably, generated by BibTEX. REVTEX 4.2 includes BibTEX style files `aparev4-2.bst`, `aparmp4-2.bst` appropriate for *Physical Review* and *Reviews of Modern Physics*, respectively.

4. Example references

This sample file employs the `\bibliography` command, which formats the `apsamp.bbl` file and specifies which bibliographic databases are to be used by BibTEX (one of these should be by arXiv convention: `apsamp.bib`). Running BibTEX (via `bibtex apsamp`) after the first pass of LATEX produces the file `apsamp.bbl` which contains the automatically formatted `\bibitem` commands (including extra markup information via `\bibinfo` and `\bibfield` commands). If not using BibTEX, you will have to create the

`thebibliography` environment and its `\bibitem` commands by hand.

Numerous examples of the use of the APS bibliographic entry types appear in the bibliography of this sample document. You can refer to the `apsamp.bib` file, and compare its information to the formatted bibliography itself.

C. Footnotes

Footnotes, produced using the `\footnote{#1}` command, usually integrated into the bibliography alongside the other entries. Numerical citation styles do this[7]; author-year citation styles place the footnote at the bottom of the text column. Note: due to the method used to place footnotes in the bibliography, *you must re-run BibTEX every time you change any of your document's footnotes.*

II. MATH AND EQUATIONS

Inline math may be typeset using the $ delimiters. Bold math symbols may be achieved using the `\mb` package and the `\bm{#1}` command it supplies. For instance, a bold α can be typeset as `$\bm{\alpha}$` giving $\bm{\alpha}$. Fraktur and Blackboard (or open face or double-struck) characters should be typeset using the `\mathfrak{#1}` and `\mathbb{#1}` commands respectively. Both are supplied by the `amssymb` package. For example, `$\mathbb{R}$` gives $\mathbb{R}$ and `$\mathfrak{G}$` gives $\mathfrak{G}$

In LATEX there are many different ways to display equations, and a few preferred ways are noted below. Displayed math will center by default. Use the class option `fleqn` to flush equations left.

Below are numbered single-line equations; this is the most common type of equation in *Physical Review*:

$$\chi_7(p) \lesssim |2\langle \mathbf{p}|\mathbf{q}\rangle + p_z|^{-1/2} \left\{ \frac{|\mathbf{p}'+p_z}{|\mathbf{p}+p_z|} \right\}. \quad (1)$$

$$\left\{ \text{ab12345678abcde123abcfp}\tfrac{1}{2}\tfrac{3}{4}\tfrac{5}{6}\text{ef} \right\}. \quad (2)$$

Note the open one in Eq. (2).

Not all numbered equations will fit within a narrow column this way. The equation number will move down automatically if it cannot fit on the same line with a one-line equation:

$$\left\{ \text{ab12345678abcde123456abcdefa}\tfrac{1}{2}\tfrac{3}{4}\tfrac{5}{6}\text{cf} \right\}. \quad (3)$$

When the `\label{#1}` command is used [cf. input for Eq. (2)], the equation can be referred to in text without knowing the equation number that TEX will assign to it. Just use `\ref{#1}`, where `#1` is the same name that used in the `\label{#1}` command.

Unnumbered single-line equations can be typeset using the \[, \] format:

$$\pi^+\pi^-\to\pi^+\pi^+\pi^-\ldots\ \pi^+\pi^-\to\pi^+\pi^+\pi^-\ldots\ .$$

A. Multiline equations

Multiline equations are obtained by using the `eqnarray` environment. Use the `\nonumber` command at the end of each line to avoid assigning a number:

$$\begin{aligned} M =& g_{\rm fi}^2(4E_1E_2)^{1/2}(\vec{v}_1^{\,\prime})^{-1}\delta_{n_-,n_+}(p'_{n_+})^2\chi_{-n_+}(p_2) \\ &\times[v_{j_1}\epsilon_{j_2j_3}\chi_{n_-}(p_1). \end{aligned} \quad (4)$$

$$\sum|M_{\rm sr}^{\rm unpol}|^2 = g_{\rm fi}^{2n-4}(Q^2)\ N^{n-2}(N^2-1) \times\left(\sum_{i<j}\right)\sum_{\rm perms}\frac{1}{S_{12}S_{34}}\frac{1}{S_{45}}\cdots\sum\mathcal{A}\ . \quad (5)$$

Note: Do not use `\label{#1}` on a line of a multiline equation if `\nonumber` is also used on that line. Incorrect cross-referencing will result. Notice the use `\pmb{#1}` for using a Roman font within a math environment.

To set a multline equation without any equation numbers, use the `\begin{eqnarray*}`, `\end{eqnarray*}` format:

$$\sum|M_{\rm sr}^{\rm unpol}|^2 = g_{\rm fi}^{2n-4}(Q^2)\ N^{n-2}(N^2-1) \times\left(\sum_{i<j}\right)\left(\sum_{\rm perms}\frac{1}{S_{12}S_{23}S_{34}}\right)\frac{1}{S_{45}}$$

To obtain numbers not normally produced by the automatic numbering, use the `\tag{#1}` command, where `#1` is the desired equation number. For example, to get an equation number of (2.6′),

$$g^+g^-\to g^+g^+g^-\ldots\ +\ g^+g^-\to g^+g^+g^-\ldots\ (2.6')$$

a. A few notes on `tags` `\tag{#1}` requires the `amsmath` package. Place the `\tag{#1}` command before the `\label{#1}`, if any. The numbering produced by `\tag{#1}` *does not affect* the automatic numbering in REVTEX; therefore, the number must be known ahead of time, and it must be manually adjusted if other equations are added. `\tag{#1}` works with both single-line and multiline equations. `\tag{#1}` should only be used in exceptional cases—do not use it to number many equations in your paper. Please note that this feature of the `amsmath` package is not compatible with the `hyperref` (6.77a) package.

Enclosing display math within `\begin{subequations}` and `\end{subequations}` will produce a set of equations that are labelled with letters, as shown in Eqs. (6b) and (6a) below. You may include any number of single-line and multiline

equations, although it is probably not a good idea to follow one display math directly after another.

$$M =& g_{\rm fi}^2(4E_1E_2)^{1/2}(\vec{v}_1^{\,\prime})^{-1}(p'_{n_+})^2\chi_{-n_+}(p_2) \\ &\times[v_{j_1}\chi_{n_-}(p_1). \quad (6a)$$

$$\left\{ \text{abc123456abcdefa}\tfrac{1}{2}\tfrac{3}{4}\tfrac{5}{6}\text{cf} \right\}. \quad (6b)$$

Giving a `\label{#1}` command directly after the `\begin{subequations}`, allows you to reference all the equations in the `subequations` environment. For example, the equations in the preceding subequations environment were Eqs. (6).

1. Wide equations

The equation that follows is set in a wide format, i.e., it spans the full page. The wide format is reserved for long equations that cannot easily be set in a single column:

$$\mathbb{R}^{(d)} = g'_{\kappa}\left(\frac{\Gamma^2(3,2)|_{2\kappa}}{Q_{12}^2-M_H^2}+\frac{\Gamma^2(3,2)|_{2\kappa}}{Q_{13}^2-M_H^2}\right)+x_W Q_e\left(\frac{\Gamma^{-1}(3,2)|_{2\kappa}}{Q_{12}^2-M_H^2}+\frac{\Gamma^{-1}(3,2)|_{2\kappa}}{Q_{13}^2-M_H^2}\right) \quad (7)$$

This is typed to show how the output appears in wide format. (Incidentally, since there is no blank line between the `equation` environment above and the start of this paragraph, this paragraph is not indented.)

III. CROSS-REFERENCING

REVTEX will automatically number such things as sections, footnotes, equations, figure captions, and table captions. In order to reference them in text, use the `\label{#1}` and `\ref{#1}` commands. To reference a particular page, use the `\pageref{#1}` command.

The `\label{#1}` should appear within the section heading, within the footnote text, within the equation, or within the table or figure caption. The `\ref{#1}` command is used in text at the point where the reference is to be displayed. Some examples: Section I on page 1, Table I, and Fig. 1.

TABLE I. A table that fits into a single column of a two-column layout. Note that REVTEX 4 adjusts the intercolumn spacing so that the table fills the entire width of the column. Table captions are numbered automatically. This table illustrates left-, center-, decimal- and right-aligned columns, along with the use of the `ruledtable` environment which sets the Scotch (double) rules above and below the alignment, per APS style.

Left*	Centered*	Decimal	Right
1	2	3.001	4
10	20	30	40
100	200	300.0	400

* Note a.
* Note b.

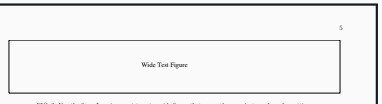


FIG. 2. Use the figure* environment to get a wide figure that spans the page to `\textwidth` formatting.

TABLE II. This is a wide table that spans the full page width in a two-column layout. It is formatted using the `table*` environment. It also demonstrates the use of `\multicolumn` in rows with entries that span more than one column.

Ion	$D^1_{\rm in}$			$D^2_{\rm in}$	
	1st alternative		2nd alternative	1st alternative	2nd alternative
K	$(3c)+(3f)$		$(3c)+(3d)$	$(3c)+(3d)$	$(3f)$
Mn	$(3p)^n$		$(3c)$	$(4s)^2$	$(3s)+(3b)$
Cl	$(a)+(b)+(c)+(d)$		$(2p)^n$	$(4s)^n$	
He	$(8c)^n$		$(4j)^n$	$(4p)^n$	
Ag			$(4k)^n$		$(4h)^n$

* The c parameter of these positions is $c = \frac{1}{8}$.

earlier in the document, as was done with Table II: LATEX's float placement algorithms need to know about a full-page-width float earlier.

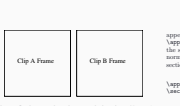
Figure 2 has content that is too wide for a single column, so the `figure*` environment has been used.

The content of a table is typically a `tabular` environment, giving rows of type in aligned columns. Column entries separated by `&`'s, and each row ends with \\. The required argument for the `tabular` environment specifies how data are aligned in the columns. For instance, entries may be centered, left-justified, right-justified, aligned on a decimal point. Extra column-spacing may be specified as well, although REVTEX 4 sets this spacing so that the columns fill the width of the table. Horizontal rules are typeset using the `\hline` command. The doubled (or Scotch) rules that appear at the top and bottom of a table can be achieved enclosing the `tabular` environment within a `ruledtabular` environment. Rows whose columns span multiple columns can be typeset using the `\multicolumn{#1}{#2}{#3}` command (for example, see the first row of Table II).

Table III. Numbers in columns Three–Five are aligned with the "d" column specifier (requires the `dcolumn` package). Nonnumeric entries (those entries without a ".") in a "d" column are aligned on the decimal point. Use the "D" specifier for more complex layouts.

One	Two	Three	Four	Five
one	two	three	four	five
He	2	3.77234	45672.	0.69
C^a	C^b	12537.64	37.66345	86.37

* Some tables require footnotes.
* Some tables need more than one footnote.



Video 1. Students explain their initial idea about Newton's third law to a teaching assistant. Clip (a): same focus. Clip (b): more backstory.

table (these footnotes will be displayed directly below the table rather than at the bottom of the page or in the bibliography). The easiest and preferred method is just to use the `\footnote{#1}` command. This will automatically enumerate the footnotes with lowercase roman letters. However, it is sometimes necessary to have multiple entries in the table share the same footnote. In this case, there is no choice but to manually create the footnotes using `\footnotemark[#1]` and `\footnotetext[#1]{#2}`. `#1` is a numeric value. Each time the same value for `#1` is used, the same mark is produced in the table. The `\footnotemark[#1]{#2}` commands are placed after the `tabular` environment. Examine the LATEX source and output for Tables I and IV for examples.

Video 1 illustrates several features new with REVTEX4.2, starting with the `video` environment, which is in the same category with `figure` and `table`. The `\westfloatlink` command causes the title of the video to be typeset in the hyperlink to the indicated URL; it may be used with any environment that takes the `\caption` command. The `\href` command has the same significance as it does in the context of the `hyperref` package; the second argument is a piece of text to be typeset in your document; the first is its hyperlink, a URL.

Physical Review style requires that the initial citation of figures or tables be in numerical order in text, so don't cite Fig. 2 until Fig. 1 has been cited.

ACKNOWLEDGMENTS

We wish to acknowledge the support of the author community in using REVTEX, offering suggestions and encouragement, testing new versions, ...

Appendix A: Appendixes

To start the appendixes, use the `\appendix` command. This signals that all following section commands refer to appendixes instead of regular sections. Therefore, the `\appendix` command should be used only once—to setup the section commands to act as appendixes. Thereafter normal section commands are used. The heading for a section can be left empty. For example,

```
\appendix
\section{}
```

will produce an appendix heading that says "APPENDIX A" and

```
\appendix
\section{Background}
```

will produce an appendix heading that says "APPENDIX A: BACKGROUND" (note that the colon is set automatically).

If there is only one appendix, then the letter "A" should not appear. This is suppressed by using the star version of the appendix command (`\appendix*` in the place of `\appendix`).

Appendix B: A little more on appendixes

Observe that this appendix was started by using

```
\section{A little more on appendixes}
```

Note the equation number in an appendix:

$$E = mc^2. \quad \text{(B1)}$$

1. A subsection in an appendix

You can use a subsection or subsubsection in an appendix. Note the numbering: we are now in Appendix B.1.

Note the equation numbers in this appendix, produced with the subequations environment:

$$E = mc, \quad \text{(B2a)}$$
$$E = mc^2, \quad \text{(B2b)}$$
$$E \gtrsim mc^3. \quad \text{(B2c)}$$

They turn out to be Eqs. (B2a), (B2b), and (B2c).

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IV. FLOATS: FIGURES, TABLES, VIDEOS, ETC.

Figures and tables are usually allowed to "float", which means that their placement is determined by LATEX, while the document is being typeset.

Use the `figure` environment for a figure, the `table` environment for a table. In each case, use the `\caption` command within to give the text of the figure or table caption along with the `\label` command to provide a key for referring to this figure or table. The typical content of a figure is an image of some kind; that of a table is an alignment.

Insert an image using either the `graphics` or `graphicx` packages, which define the `\includegraphics{#1}` command. (The two packages differ in respect of the optional arguments used to specify the orientation, scaling, and translation of the image.) To create an alignment, use the `tabular` environment.

The best place to locate the `figure` or `table` environment is immediately following its first reference in text; this sample document illustrates this practice for Fig. 1, which shows a figure that is small enough to fit in a single column.

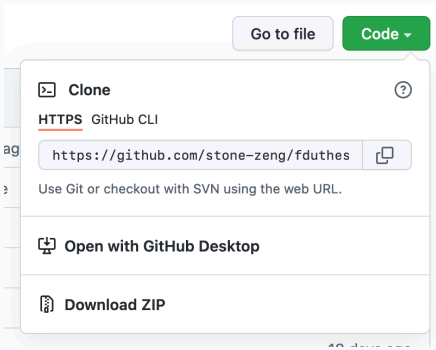
In exceptional cases, you will need to move the float



FIG. 1. A figure caption. The figure captions are automatically numbered.

fduthesis：安装

- 最新版本：v0.8-dev
- 建议直接从 GitHub 下载 🔗
  - Code > Download ZIP
  - 或使用 `git clone` / `gh repo clone`
  - 执行 `install-win.bat` 或 `install-linux.sh`，所需文件会在 `thesis` 文件夹中生成
  - Releases 中的版本较旧
- TeX Live 和 Overleaf 🔗 上的也较旧
  - ~~主要怪我懒~~
  - 会尽快更新的（



fduthesis：使用

- 文档类选项
  - 论文类型：`type = doctor|master|bachelor`
  - 单双面模式：`oneside` 或 `twoside`
- 参数设置举例

```
\fdusetup{
  style = {font-size = 5, bib-resource = {ref.bib}},
  info/title = {论动体的电动力学},
  info/author = {阿尔伯特・爱因斯坦},
}
```

- 文献引用
  - `\cite`、`\citet`、`\parencite` 等
  - 文献列表：`\printbibliography`
- 编译
  - 推荐 XeLaTeX，也可 `latexmk -pdfxe`

fduthesis：示例



学校代码：10246
学　　号：18110190000

復旦大學

博士学位论文
（学术学位）

论文标题

Thesis Title

院　　系：　物理系
专　　业：　物理学
姓　　名：　王二
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摘　要

量子力学是描述微观物质（原子、亚原子粒子）行为的物理学理论，量子力学是我们理解除万有引力之外的所有基本力（电磁相互作用，强相互作用，弱相互作用）的基础。

量子力学是许多物理学分支的基础，包括电磁学、粒子物理、凝聚态物理以及宇宙学的部分内容，量子力学也是化学键理论、结构生物学以及电子学等学科的基础。

量子力学主要适用来描述微观下的行为，所描述的粒子观象无法精确地以经典力学论释，例如：根据海哈根诠释，一个粒子在被观测之前，不具有任何物理性质，然而被观测之后，依测量仪器而定，可能观测到其粒子性质，也可能观测到其波动性质，或者观测到一部分粒子性质一部分波动性质，此即波粒二象性。

量子力学始于 20 世纪初马克斯·普朗克和尼尔斯·玻尔的开创性工作，马克斯·玻恩于 1924 年创造了“量子力学”一词。因其成功的解释了经典力学无法解释的实验现象，并精确地预言了此后的一些发现，物理学界开始广泛接受这个新理论，量子力学早期的一个主要成就是成功地解释了波粒二象性。此术语源于亚原子粒子同时表现出粒子和波的特性。

关键字：不确定关系；量子力学；理论物理
中图分类号：O413.1

Abstract

Quantum mechanics is a fundamental theory in physics that provides a description of the physical properties of nature at the scale of atoms and subatomic particles. It is the foundation of all quantum physics including quantum chemistry, quantum field theory, quantum technology, and quantum information science.

Classical physics, the collection of theories that existed before the advent of quantum mechanics, describes many aspects of nature at an ordinary (macroscopic) scale, but is not sufficient for describing them at small (atomic and subatomic) scales. Most theories in classical physics can be derived from quantum mechanics as an approximation valid at large (macroscopic) scale.

Quantum mechanics differs from classical physics in that energy, momentum, angular momentum, and other quantities of a bound system are restricted to discrete values (quantization), objects have characteristics of both particles and waves (wave–particle duality), and there are limits to how accurately the value of a physical quantity can be predicted prior to its measurement, given a complete set of initial conditions (the uncertainty principle).

Quantum mechanics arose gradually from theories to explain observations which could not be reconciled with classical physics, such as Max Planck's solution in 1900 to the black-body radiation problem, and the correspondence between energy and frequency in Albert Einstein's 1905 paper which explained the photoelectric effect. These early attempts to understand microscopic phenomena, now known as the "old quantum theory", led to the full development of quantum mechanics in the mid-1920s by Niels Bohr, Erwin Schrödinger, Werner Heisenberg, Max Born and others. The modern theory is formulated in various specially developed mathematical formalisms. In one of them, a mathematical entity called the wave function provides information, in the form of probability amplitudes, about what measurements of a particle's energy, momentum, and other physical properties may yield.

Keywords: Uncertainty principle; quantum mechanics; theoretical physics
CLC code: O413.1

第 1 章　数学基础

量子力学是物理学的分支学科。它主要描写微观的事物，与相对论一起被认为是现代物理学的两大基本支柱，许多物理学理论和科学，如原子物理学，固体物理学，核物理学和粒子物理学以及其它相关的学科，都是以其为基础[1–3]。

1.1　基础公设

整个量子力学的数学理论可以建立于五个基础公设。这些公设不能被严格推导出来的，而是从实验结果仔细分析归纳总结而得到的，从这五个公设，可以推导出整个量子力学。假若量子力学的理论结果不符合实验结果，则必须将这些基础公设加以修改，直到没有任何不符合之处。至今为止，量子力学已被实验核对至极高准确度，还没有找到任何与理论不符合的实验结果，虽然有些理论很难直觉地用经典物理的概念来理解，例如，波粒二象性，量子纠缠等等[4–10]。

1. 量子态公设：量子系统在任意时刻的状态（量子态）可以由希尔伯特空间 $\mathscr{H}$ 中的态矢量 $|\psi\rangle$ 来设定，这态矢量完备地给出了这量子系统的所有信息，这公设意味着量子系统遵守生命如常理，假若 $|\psi_1\rangle$、$|\psi_2\rangle$ 属于希尔伯特空间 $\mathscr{H}$，则 $c_1|\psi_1\rangle + c_2|\psi_2\rangle$ 也属于希尔伯特空间 $\mathscr{H}$。

2. 时间演化公设：态矢量为 $|\psi(t)\rangle$ 的量子系统，其动力学演化可以用薛定谔方程表示：

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle\,. \tag{1.1}$$

其中，哈密顿算符 $\hat{H}$ 对应于量子系统的总能量，$\hbar$ 是约化普朗克常数，根据薛定谔方程，假设时间从 t_0 变化到 t，则态矢量从 $|\psi(t_0)\rangle$ 演化到 $|\psi(t)\rangle$，该过程以方程表示为

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,. \tag{1.2}$$

其中 $\hat{U}(t,t_0) = \mathrm{e}^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}$ 是时间演化算符。

3. 可观察量公设：每个可观察量 A 都有其对应的厄米算符 $\hat{A}$，而算符 $\hat{A}$ 的所有本征矢量共同组成一个完备基底。

4. 坍缩公设：对于量子系统测量某个可观察量 A 的过程，可以数学表示为将对应的厄米算符 $\hat{A}$ 作用于量子系统的态矢量 $|\psi\rangle$。测量值只能为厄米算符 $\hat{A}$ 的本征值，在测量后，假设测量值为 a_i，则量子系统的量子态立刻会坍塌为对应于本征值 a_i 的本征态 $|e_i\rangle$。

5. 波恩公设：对于这测量，获得本征值 a_i 的概率为量子态 $|\psi\rangle$ 处于本征态 $|e_i\rangle$ 的概率幅的绝对值平方。[1]

1.2　量子态与量子算符

量子态指的是量子系统的状态，态矢量可以用来抽象地表现量子态。采用狄拉克标记，态矢量表示为右矢 $|\psi\rangle$；其中，在符号内部的希腊字母 ψ 可以是任何符号，字母，数字，或单字，例如，沿着磁场方向测量电子的自旋，得到的结果可以是上旋或是下旋，分别标记为 $|\uparrow\rangle$ 和 $|\downarrow\rangle$。



图 1-1　设定施特恩－格拉赫实验仪器的磁场方向为 z-轴，入射的银原子束可以被分裂成两道银原子束，每一道银原子束代表一种量子态，上旋 $|\uparrow\rangle$ 或下旋 $|\downarrow\rangle$[10]。

对量子态做操作定义，量子态可以从一系列制备程序来辨认，即这程序所制成的量子系统拥有这量子态，例如，使用施特恩－格拉赫实验仪器，设定磁场朝着 z-轴方向，如图 1-1 所示，可以得入射的银原子束，依照自旋的 z-分量分裂成两道，一道为上旋，量子态为 $|\uparrow\rangle$；另一道为下旋，量子态为 $|\downarrow\rangle$，这样，可以制备成量子态为 $|\uparrow\rangle$ 的银原子束，或量子态为 $|\downarrow\rangle$ 的银原子束。原本银原子束的态矢量可以按照态叠加原理表示为

$$|\psi\rangle = \alpha\,|\uparrow\rangle + \beta\,|\downarrow\rangle\,. \tag{1.3}$$

其中，α、β 是复值系数，$|\alpha|^2$、$|\beta|^2$ 分别为入射银原子束处于上旋、下旋的概率，且有

$$|\alpha|^2 + |\beta|^2 = 1. \tag{1.4}$$

1.3　动力学演化

量子系统的动力学演化可以用不同的绘景来表现。通过重新定义，这些不同的绘景可以互相变换，它们实际上是等价的。假若要专注分析量子态怎样随着时间的流易而演化，时间演化算符怎样影响量子态，则可采用薛定谔绘景。假若要专注了解对应于可观察量的算符怎样随着时间的流易而演化，时间演化算符怎样影响这些算符，则可采用海森堡绘景。

1　假设可观察量 A 的基底 $\{e_1, e_2, \ldots, e_n\}$，量子态 $|\psi\rangle$ 可以表示为 $|\psi\rangle = \sum_i c_i\,|e_i\rangle$，其中 c_i 是量子态 $|\psi\rangle$ 处于本征态 $|e_i\rangle$ 的概率幅，根据波恩证明，对于此次测量，获得本征值 a_i 的概率为 $|\langle e_i|\psi\rangle|^2 = |c_i|^2$。

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叁拾伍

公式

数学模式

- 一切数学公式都要在数学模式下输入
  - 不受外界字体命令控制
  - 数学模式中空格不起作用，尽管用；但不能有空行
  - 建议始终调用 `amsmath` 宏包
  - **不建议用 MathType 生成 LaTeX 公式**
  - 但可以用 MathJax 或 KaTeX 练习
- 行内（inline）公式
  - 用一对美元符号（公式值千金）：`$...$`
  - 示例：理想气体状态方程可以写为 $PV = nRT$，其中 P、V 和 T 分别是压强、体积和绝对温度
- 独显（display）公式
  - 无编号：`\[...\]` 或 `equation*` 环境
  - 编号：`equation` 环境
  - **不要用 `$$...$$`**

结构

- 上下标
  - `^` 和 `_`：`f^ab` 和 `f^{ab}`，`e^x^2`、`{e^x}^2` 和 `e^{x^2}`
  - 张量：`R^a{}_b{}^{cd}` 或使用 tensor 宏包
  - 配合积分、求和、极限使用：`\int`、`\sum`、`\lim`；`\(no)limits`
- 分式
  - `\frac{<分子>}{<分母>}`
  - 行内分式、小分式不好看：改用 `a/b`，或改用独显公式
  - **不推荐 `\dfrac`**
- 根式
  - `\sqrt[<次数>]{<内容>}`
  - 复杂情况改用分数指数：`{...}^{1/n}`
- 矩阵与行列式
  - `matrix`、`pmatrix`、`vmatrix` 等环境
  - 语法类似表格：`&` 分列，`\\` 换行
  - 推荐 physics 宏包

括号与定界符

- 基本括号
  - `(...)`、`[...]`、`\{...\}`、
  - 绝对值、范数：`|...|` 或 `\vert...\vert`、`\Vert...\Vert`
  - Dirac 符号：`\langle...\rangle`、`|...\rangle`
- 自动调节
  - `\left(...\right)` 等
  - 大型括号是拼出来的
- 手动调节
  - 只有 4 + 1 档：`\big`、`\Big`、`\bigg`、`\Bigg`
  - 声明左中右：`\bigl`、`\bigm`、`\bigr` 等

符号与字体

- 符号不是按钮点出来的，也不是天上掉下来的
  - （几乎）所有的符号都由字体提供
  - 分清「它是什么」和「它长什么样」（术语：character 和 glyph）
- 寻找符号
  - 最常用的额外字体包：`amssymb`
  - S. Pakin. *The Comprehensive LaTeX Symbol List* 🔗
  - 手写识别（有趣但不全）：Detexify 🔗
- 数学字体
  - 你们要的「Times New Roman」：`newtxmath` 宏包
  - **不要用 `times` 和 `mathptmx` 宏包**
  - 加粗：使用 `bm` 宏包的 `\bm` 命令（`\mathbf` 只有直立的字母）
- 新方案：`unicode-math` 🔗
  - 符号、字体、样式精调的一揽子解决方案
  - 彻底修改底层，不可与传统方案混用

多行公式

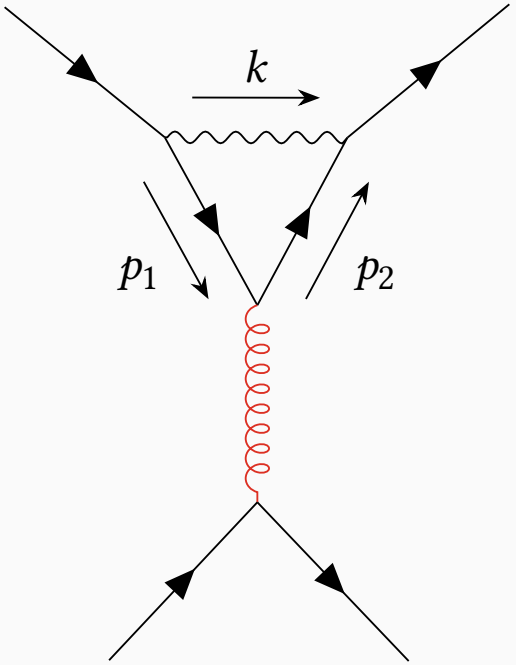
- 以下均要求 amsmath 宏包
- 独立数学环境
  - 多行居中 gather、多行手动对齐 align、跨行 multiline
  - 手动对齐：关系符前加 &
- 内联数学环境
  - 条件 cases、多行对齐 split、...ed
- 精细调整
  - mathtools、empheq 等
  - 自动换行：breqn
  - **避免使用 eqnarray 环境**

专业功能（一）

- 更高更妙的物理：physics 宏包
  - 括号：`\qty(...)`、`\qty\big{...}`
  - 矩阵：`\mqty(...)`、`\mqty|...|`、`\dmat{a,b,c,...}`
  - Dirac 符号：`\ket`、`\bra`、`\ev`
  - 向量、导数、微分、更多函数名……
- 国际单位：siunitx 宏包
  - `$4.18 \times 10^3 J mol^{-1} K^{-1}$`
    - $4.18 \times 10^3 Jmol^{-1}K^{-1}$—No!
  - `\qty{4.18e3}{J.mol^{-1}.K^{-1}}`
    - $4.18 \times 10^3 \mathrm{J\,mol^{-1}\,K^{-1}}$—Yes!
  - 注 1：此宏包代码比 LaTeX 2_ε 内核还长
  - 注 2：新版本的 siunitx 宏包与 physics 宏包有兼容性问题

专业功能（二）

- 花式图表
  - Feynman 图：`tikz-feynman` 宏包 (arXiv: 1601.05437 🔗)
  - Feynman 斜线：`slashed` 宏包
  - Wick 定理：`simplewick` 宏包、`simpler-wick` 宏包
  - Young 表、Young 图：`ytableau` 宏包
  - 电路图：`circuitikz` 宏包
  - 拓扑量子场论：`tqft` 宏包
  - ……



专业功能（三）

- 抄录：忽略所有特殊类别码（catcode），原样显示
  - `\verb<char>...<char>`、`verbatim` 环境
  - `verbatim`、`fancyvrb` 宏包
- 语法高亮
  - `listings` 宏包
  - `minted` 宏包
    - 需要 Python，且开启 `--shell-escape`

```c
/* A standard Hello World program in C. */
#include <stdio.h>

int main(int argc, char** argv) {
    printf("Hello, world!\n");
    return 0;
}
```

字体排印

先看一个例子

Typography is the art and technique of arranging type to make written language legible, readable, and appealing when displayed. The arrangement of type involves selecting typefaces, point sizes, line lengths, line-spacing(*leading*), and letter-spacing (*tracking*), and adjusting the space between pairs of letters(*kerning*). The term typography is also applied to the style, arrangement, and appearance of the letters, numbers, and symbols created by the process. **Type design** is a closely related craft, sometimes considered part of typography;most typographers do not design typefaces, and some type designers do not consider themselves typographers. Typography also may be used as a decorative device, unrelated to communication of information.

中易黑体，行距 0.8 倍，关闭断词；文本来源：🔗

没有对比就没有伤害

Typography is the art and technique of arranging type to make written language legible, readable, and appealing when displayed. The arrangement of type involves selecting typefaces, point sizes, line lengths, line-spacing (*leading*), and letter-spacing (*tracking*), and adjusting the space between pairs of letters (*kerning*). The term typography is also applied to the style, arrangement, and appearance of the letters, numbers, and symbols created by the process. **Type design** is a closely related craft, sometimes considered part of typography; most typographers do not design typefaces, and some type designers do not consider themselves typographers. Typography also may be used as a decorative device, unrelated to communication of information.

EB Garamond，默认设置

术语

- 语言（language）
- 文字（script）
- 书写系统（writting system）
- 符号（symbol）
- 字符（character）
- 字符形（glyph）
- 字符集（character set）
- 编码（encoding）
- 码位（code point）
- 字体（font）
- 字型（typeface）
- 易认性（legibility）
- 可读性（readability）
- 字偶间距（kerning）
- 字距（tracking）
- 栅格化（rasterization）
- 渲染提示（hinting）
- ……

LaTeX will do (almost) all the things for you.

Punctuations: hyphen/dash

- Hyphen (U+002D): `-`
  - Four-dimensional momentum
  - Hyphenation
    - Hyphenation is language-dependent
    - Use a dictionary to check breakable positions
- En dash (U+2013): `--`
  - Ryu–Takayanagi formula (*cf.* Levi-Civita symbol)
  - pp. 187–189
- Em dash (U+2014): `---`
  - Red, white, and blue—these are the colors of the flag
  - Like colon, parentheses, *etc.*
- Minus (U+2212): `$-$`
  - $a - b$, $-a$

A hyphen-ation algo-rithm is a set of rules, especially one codified for imple-mentation in a computer program, that decides at which points a word can be broken over two lines with a hyphen.

Punctuations: quotation mark

- Left/right, single/double:
  - ‘...’ (`U+2018`, `U+2019`): `` `...' ``
  - “...” (`U+201C`, `U+201D`): `` ``...'' ``
- Different languages:
  - ‘British “English” style’ and “American ‘English’ style”
  - „German”, ”Finnish”, «French», »Danish«, *etc.*
  - Use `csquotes` package
- Programming:
  - `char* my_name = "Xiangdong Zeng";`
- Mathematics:
  - `f' = f^{\prime}`: $f' = f^{\prime}$

中文标点符号

- 句号
  - 正常文本。科技文本.
- 引号
  - 『传统风格』，「某乎风格」，“标准风格”，’奇葩风格”
- 破折号
  - 断开——是不好的，不断开——是好的
- 波浪号：
  - `~` ≠ `\textasciitilde` ≠ `\texttildelow` ≠ `$\sim$` ≠ 你要的那个
  - 那就是 青~藏˜高~原～～～～
    - U+007E: Tilde
    - U+02F7: Modifier letter low tilde
    - U+223C: Tilde operator
    - U+FF5E: Fullwidth tilde
    - ...

使用字体：fontspec 宏包

- 字体家族 🔗
  - 衬线体：EB Garamond, Times New Roman, Latin Modern, *etc.*
  - 无衬线体：Helvetica Neue, Avenir Next, Optima, *etc.*
  - 等宽字体：Courier, Menlo, GNU Unifont, *etc.*
  - 中文字体：宋体、**黑体**、仿宋、楷书、（娃娃体）……
- 样式 🔗
  - 粗体、意大利体：**Bold** vs **Faked bold**, *Italic* vs *Slant*
  - **汉字一般不使用斜体**
  - 视觉字号（optical size）：

    Tiny, Script, Footnote, Caption, Small, Normal, Large, Huge
- OpenType 特性
  - 连字（ligature）：ff → ff, fi → fi, fl → fl
  - 老式数字（old-style number）：0123456789 → 0123456789
  - 字偶间距（kerning）：Ty → Ty, WA → WA
- **请避免滥用过多字体**（此页除外）

进阶扩展

TeX 宏编程

- TeX 层面
  - 守序善良——定义命令：`\def`、`\gdef`、`\let`
  - 绝对中立——展开控制：`\edef`、`\expandafter`、`\aftergroup`
  - 混乱邪恶——类别码：`\catcode`
- LaTeX 层面
  - 定义新命令：`\newcommand`、`\renewcommand`、`\newenvironment`
  - 内部命令：`\makeatletter`、`\makeatother`
- LaTeX3——可望还不可即的未来 🔗 🔗
  - 命令举例：`\cs_new:cpx`、`\seq_sort:Nn`、`\regex_match:nnTF`
  - 创建用户层命令：xparse 宏包（已默认载入）
  - 面向对象编程、组件化：xtemplate 宏包
  - fontspec、siunitx、ctex、fduthesis 等均使用 LaTeX3 实现
  - LaTeX 内核正逐步迁移至 LaTeX3
- 外部语言调用
  - `\write18`、`\directlua` 与 PythonTeX

深入字体

- NFSS 与字体的坐标
  - 字体族、形状、系列、编码、字号
  - TEX 仅需要 metric 信息：`.tfm`
- 现代方案：OpenType
  - 编码层面：支持 Unicode
  - 东亚文字：超大字符集、地区变体、竖排
  - 中东、南亚文字：Bi-di 文本、上下文连字、字符序调整
- OpenType 中的数学支持（`MATH` 表）
  - Unicode Math：字母、符号支持
  - 「数学常数」：整体度量信息——上下标位置、分数线粗细等
  - `MathVariants`：大小替换（积分号、根号、括号等）
  - `MathGlyphConstruction`：字符形装配（更大的根号、括号等）
- 可变字体（variable **font**）、絵文字（emoji, 😂）……

编写宏包

- 文学编程
  - 代码、注释与文档合为一体（`.dtx` 文件）
  - 使用 `doc` 与 `docstrip` 宏包
  - 但也增加了额外的复杂度
- 发布
  - 上传 CTAN 实际上并无门槛
  - 但仍有必要了解：
    - TEX 目录结构（TDS）
    - 测试系统：`l3build` 宏包
    - 版本控制、持续集成
    - 许可证选择：LATEX 内核使用 LPPL 1.3c 🔗
- 参考
  - LATEX 内核代码：`source2e.pdf`、`classes.pdf`、`source3.pdf`
  - 书籍：*The TEXbook*、*TEX by Topic*、*The LATEX Companion* 等

宏包推荐

- 必备
  - amsmath
  - graphicx
  - hyperref
- 样式
  - caption
  - enumitem
  - fancyhdr
  - footmisc
  - geometry
  - ntheorem
  - titlesec
- 数学
  - bm
  - mathtools
  - physics
  - unicode-math
- 表格
  - array
  - booktabs
  - longtable
  - tabularx
- 插图、绘图
  - float
  - pdfpages
  - standalone
  - subfig
  - pgf/tikz
  - pgfplots
- 字体
  - newtx
  - newpx
  - pifont
  - fontspec
- 多语言
  - babel
  - polyglossia
  - ctex
  - xeCJK
  - luatexja
- 杂项功能
  - algorithm2e
  - beamer
  - biblatex
  - fancyhdr
  - listings
  - mhchem
  - microtype
  - minted
  - natbib
  - siunitx
  - xcolor

请务必先读文档!

命令行执行 texdoc *package*

Markdown

```
# Markdown syntax

This is **bold text**.
This text is *italicized*.
Use `git status` to list all
new or modified files.

Block code:

```
git status
git add
git commit
```

Quotation:

> Markdown uses email-style `>`
> characters for blockquoting.

List

Bullet list

* apples
* oranges
* pears

Numbered list

1. wash
2. rinse
3. repeat

Link: from [Wikipedia]
(https://en.wikipedia.org/wiki/
Markdown)
```

Git

- 版本管理的必要性
  - 远离「初稿，第二稿，第三稿……终稿，终稿（打死也不改了）」
  - 有底气做大范围修改、重构
  - 方便与他人协同合作
- 基本用法
  - 把大象放进冰箱：`git init`、`git add`、`git commit`
  - 时空穿梭：`git reset`、`git revert`
  - 平行宇宙：`git branch`、`git checkout`、`git rebase`
  - 推荐用 VS Code 等进行可视化操作
  - 参考链接：🔗🔗
- GitHub
  - 远程 Git 仓库
  - Clone & fork
  - Issues & pull requests
  - **提醒：绑定 `.edu` 邮箱可以有更多优惠**

获取帮助

- 搜索、提问的姿势
  - 优先使用英文 + Google (if possible)
  - 提供最小工作示例（MWE, minimal working example）
    - 能复现问题
    - 尽量不带冗余内容
    - 策略：二分查找
  - 遵循社区行为准则（code of conduct）
- 在线论坛
  - TeX - LaTeX Stack Exchange 🔗
  - CTeX 临时论坛 🔗
  - LaTeX 工作室 🔗
    - 资源需要甄别，且部分内容需付费

社区参与

- 文档翻译
  - `lshort-zh-cn` 🔗
  - `learnlatex.org/zh` 🔗
- 宏包开发与维护
  - 不妨先从修 typo 开始
  - 参与讨论，你的经验也可以解他人之忧 🔗 🔗 🔗
  - 欢迎参与维护 `fdutheis`
- ~~来当主讲人~~

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关于

本幻灯片：https://github.com/stone-zeng/latex-talk
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Beamer 主题：萧山
正文字体：思源宋体 + Libertinus Serif
等宽字体：思源黑体 + Roboto Mono

\bye